Bayesian t-Test Sample Size Determination: Reference Tables for Various Bayes Factor Thresholds, Effect Sizes, Sample Sizes, and Variance Assumptions

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Draft Version 2.0

17 November 2022

This pre-print version has not undergone peer-review.

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Abstract

Psychologists and other social scientists are often interested in statistically testing for differences between two means, with the use of a *t*-test. Recently, Bayesian methods for statistical testing have become more popular amongst psychologists. Bayesian methods, including the Bayesian *t*-test result in a ratio, called a Bayes Factor (*BF*), which indicates the likelihood of one model over another (commonly the alternative vs the null hypothesis models). The *BF* is an updatable ratio, which increases in favor of the most probable model as data is accumulated. Therefore, the *BF* enables researchers to decide whether evidence is compelling enough to stop with data collection, or if more observations should be obtained. However, data collection cannot always rely on this updatable property of the *BF* and sample size determination is often required a-priori. Determining the required sample size within the Bayesian framework, depends upon simulations, which can be timely and complicated to implement. Therefore, we conducted 1600 simulations with 10000 iterations each, to provide sample size determination heuristics for researchers and other stakeholders based on common assumptions expected within the social sciences. We summarize these heuristics in eight reference tables, which provide the probability of reaching five different *BF* thresholds for a Bayesian *t*-test, for evidence in favor of both the alternative and the null hypotheses, for 10 possible effect sizes, eight sample sizes, and four different data variability assumptions.

Keywords: Bayesian, t-test, sample size, simulation, Monte Carlo

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Researchers in psychology and the broader social sciences are often interested in comparing two means to study a phenomenon. To statistically analyze such comparisons, psychologists often relied on conventional null-hypothesis significance testing, by utilizing -most commonly- the Student's *t*-test (Delacre et al., 2017), which results in a *p*-value indicating whether a null hypothesis can be rejected or not (for a critical review of this approach see Scheel et al., 2021). However, during the past decade, psychology as a science has received criticisms regarding its reproducibility and rigorousness, with the traditional null-hypothesis significance testing approach having a crucial role in this critique (e.g., Derksen, 2019; Nelson et al., 2018; Scheel, 2022; Scheel et al., 2021). Fortunately, these criticisms have facilitated encouraging advancements in the field of psychology, such as the introduction of preregistered reports (the process where study proposals are peer reviewed and pre-accepted, based on the theory and methods, before research is undertaken; Chambers et al., 2015; Chambers & Tzavella, 2022; Nelson et al., 2018; Nosek et al., 2018) and the promotion of Bayesian statistics (Dienes, 2014, 2019, 2021a, 2021b; Dienes & Mclatchie, 2018; Scheel et al., 2021).

Within the Bayesian framework, statistical models are implemented, which provide a likelihood ratio called the Bayes Factor (*BF*). The *BF* indicates how likely it is for the observed data to stem from one of two competing models; most commonly the alternative and the null hypothesis (usually denoted "BF_{10}" and, respectively, "BF_{01}" to signify the ratio of the null hypothesis over the alternative hypothesis) (Derksen, 2019; Dienes, 2014, 2019, 2021a, 2021b; Dienes & Mclatchie, 2018; Rouder et al., 2009; Wagenmakers et al., 2010). As an example, a $BF_{10} = 2$, indicates that the data are two times more likely to have been observed under the alternative hypothesis, whereas a $BF_{10} = 0.5$ shows that the data are two

times more probable under the null hypothesis. Regarding the Bayesian *t*-test specifically, Rouder and colleagues (2009) provide a thorough description (see also Kruschke, 2013; see Wagenmakers et al., 2010 for an introduction to the Savage-Dickey method; see Fu et al., 2021 for an introduction to the approximate adjusted fractional *BF*). Because of this advantage of Bayesian statistics that allow inferences about the null hypothesis, as opposed to traditional frequentist analyses, a Bayesian approach is often preferred for psychological research (Dienes & Mclatchie, 2018; Kruschke, 2013; Schönbrodt & Wagenmakers, 2018; van Ravenzwaaij & Etz, 2021; Wagenmakers et al., 2010), and especially for registered reports (Chambers et al., 2015; Chambers & Tzavella, 2022; Nelson et al., 2018; Nosek et al., 2018).

Further, the *BF* is an updatable ratio (hence it is often referred to as the *predictive updating factor;* Schönbrodt & Wagenmakers, 2018), which means that as additional data are accumulated, the evidence in favor of the most probable hypothesis, reflected by the *BF*, increases (Fu et al., 2021; Schönbrodt et al., 2017; Schönbrodt & Wagenmakers, 2018; van Ravenzwaaij & Etz, 2021; Wagenmakers et al., 2010). Therefore, researchers are not necessarily depended upon an a-priori determined sample size, but can continue collecting data, until enough evidence has been obtained (i.e., when the *BF* reaches a predefined threshold) (Fu et al., 2021; Schönbrodt et al., 2017; Schönbrodt & Wagenmakers, 2018; van Ravenzwaaij & Etz, 2021; Wagenmakers et al., 2010). Despite this updatable property of the *BF*, researchers are often required to define or estimate their anticipated sample size a-priori (e.g., Fu et al., 2021; Schönbrodt & Wagenmakers, 2018); for example, when preparing research proposals, for estimating resource efficacy, or for study pre-registration (see Chambers & Tzavella, 2022). Unfortunately, determining the appropriate sample size for a Bayesian analysis, is not always straightforward and it often entails sophisticated simulations (Fu et al., 2021; van Ravenzwaaij & Etz, 2021; see also Kruschke, 2013; Schönbrodt et al.,

2017; Schönbrodt & Wagenmakers, 2018), which can be timely and complicated to implement.

Previous work has discussed estimations of minimum required sample sizes in Bayesian designs (Brysbaert, 2019; Palfi & Dienes, 2019; Schönbrodt et al., 2017). However, these estimations rely on qualitative assessments and often rest on assumptions that cannot reflect many common anticipated designs. Further, previous simulations have been conducted to provide guidance for the Bayesian *t*-test and Welsch test (Fu et al., 2021), however these simulations were designed using the approximate adjusted fractional *BF* (Gu et al., 2018) approach, and only for a specific prior distribution (Rouder et al., 2009). Even though the approximate adjusted fractional *BF* is implemented in various statistical packages (e.g., *BayesFactor* package; Morrey et al., 2015), statistical software that is commonly preferred by researchers for conducting Bayesian analyses, such as JASP (http://www.jasp-stats.org), depend upon the Savage-Dickey method (Wagenmakers et al., 2010) to calculate a *BF* (see Rouder et al., 2009; van Doorn et al., 2021; Wagenmakers et al., 2018). Consequently, psychologists currently lack specified heuristics to determine their sample size for Bayesian designs interested in testing differences between two means. Therefore, the aim of this study, is to provide reference tables for sample size justification, through Monte Carlo simulations, for researchers and other stakeholders (reviewers, editors, funders, etc.) who are interested in testing differences between two means using a Bayesian *t*-test. These heuristics are provided for numerous *BF* thresholds, for various effect size expectations, for several sample sizes, and for different assumptions regarding the data variance.

Simulation Procedure

Monte Carlo simulations were implemented in Python (version 3.9.13) and all statistical analyses were computed using the Pingouin package (Vallat, 2018). Monte Carlo simulations were conducted to estimate the probability of reaching or surpassing a specified

BF amongst five different thresholds, for 10 potential effect sizes, eight possible sample sizes, and four different assumptions regarding the sample variance (see Figure 1). These combinations resulted in 1600 separate simulations, which were based on a previously used simulation, that was applied to justify a-priori the sample size of two different experiments that each compared two means with a Bayesian *t*-test (Phylactou et al., 2022; but see also Kruschke, 2013; Schönbrodt & Wagenmakers, 2018 for similar simulation designs). Specifically, each simulation compared the data derived from two normal distributions, using a Bayesian paired *t*-test, to generate a *BF*. The prior distribution for the Bayesian paired *t*-test was described by a Cauchy distribution centered on 0, which is commonly preferred for psychological research (Rouder et al., 2009), with its scale *r* equal to the expected effect size in each simulation (see below). Independent simulations were conducted for $j = 10$ different effect sizes (0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1), which correspond to a range of standardized effect sizes (i.e., Cohen's *d*) that are commonly anticipated in the social sciences and psychology (for a discussion on the topic see Brysbaert, 2019). In each simulation $k =$ 10000 iterations were computed, resulting in 10000 *BFs* for each of the 1600 simulations.

Simulations were conducted to calculate the probability of surpassing *BF* thresholds of 2, 3, 5, 6, 10, which are common thresholds of interest within psychological research (Brysbaert, 2019; Palfi & Dienes, 2020; Schönbrodt et al., 2017; Wagenmakers et al., 2010). Separate simulations were conducted to estimate the probability of reaching the *BF* threshold for when a difference approximating the anticipated effect size was expected and for when a null effect (i.e., no difference) was expected. To calculate the probability of *BF* when a difference is expected, the first normal distribution drew it's mean from the Cauchy distribution that was used as a prior distribution to inform the Bayesian *t*-test (Figure 1A), were *r* was set to equal to the expected effect size ($\mu^1 \sim \text{Cauchy}[0, r]$). The second normal distribution had a mean equal to 0 ($\mu^2 = 0$). In the case of calculating the probability of *BF*

when no difference was expected both normal distributions had a mean equal to 0 ($\mu^1 = \mu^2 = 0$). In all simulations, the variance of each normal distribution, reflected by its standard deviation, was drawn from a uniform distribution ($sd \sim \text{Uniform}[\alpha, \beta]$). The parameters of these uniform distributions [α, β] were set according to $i = 4$ different assumptions, resulting in separate simulations for data with low variability ($i_{low} \sim \text{Uniform}[0, 1]$), moderate variability ($i_{moderate} \sim \text{Uniform}[1, 2]$), large variability ($i_{large} \sim \text{Uniform}[2, 3]$), and uninformative variance ($i_{uninformative} \sim \text{Uniform}[0, 4]$). Finally, the number of observations drawn from each normal distribution varied, so that independent simulations were conducted for eight different sample sizes (5, 10, 20, 30, 40, 50, 75, 100). Notably, each sample size echoes the number of comparisons in the Bayesian *t*-test. Therefore, each sample size corresponds to the total sample size for a within-subject dependent (paired) *t*-test, and to the single group size for a between-subject independent *t*-test, assuming that the two groups have equal sizes (thus the total sample size for an independent *t*-test should be estimated as two times the simulated sample size).

In summary, the 1600 independent simulations that were conducted, reflected assumptions for the Bayesian *t*-test, which are commonly discussed in the psychological and broader social science fields, regarding the prior distribution of the *t*-test (Rouder et al., 2009), the expected effect sizes (see Brysbaert, 2019), and the *BF* threshold (Palfi & Dienes, 2020; Schönbrodt et al., 2017; Wagenmakers et al., 2010). Further, to offer various heuristics to social science researchers, the simulations were repeated for eight different sample sizes, and for four different variability assumptions. These heuristics, which were summarized in eight reference tables, are discussed in the following section.

Reference Tables

The 1600 Monte Carlo simulations were summarized in eight different tables to be used as reference. Odd numbered tables (Tables 1, 3, 5, 7) involve simulations generated to

calculate the probabilities of exceeding the *BF* threshold if the expected effect size is true (evidence in favor of the alternative hypothesis), while tables with even numbers (Tables 2, 4, 6, 8) concern the probabilities of exceeding the *BF* threshold, when no difference is expected and thus the true difference is null (evidence in favor of the null hypothesis). Tables 1 and 2 provide the probabilities of surpassing the *BF* thresholds for data with unknown variability (uninformative variance prior), while Tables 3 and 4 for data with low variability, Tables 5 and 6 for moderate variability, and Tables 7 and 8 for high variability. Within each table, the percentage of reaching or surpassing five different *BF* thresholds (2, 3, 5, 6, 10) for eight different sample sizes (5, 10, 20, 30, 40, 50, 75, 100), and 10 different effect sizes (0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1) are provided.

Notably, as is apparent by the reference tables, evidence in favor of the null hypothesis (i.e., no difference) is much less probable to be obtained compared to evidence in favor of the alternative hypothesis for smaller effect sizes and higher *BF* thresholds. This is because the value 0, which represents the null effect (i.e., no difference) in these simulations, is at the highest density point of the prior distribution, which requires evidence to a greater extent compared to any other value to reach and exceed the *BF* threshold (Wagenmakers et al., 2010). Because of this and given that the *BF* is an updatable ratio (Schönbrodt & Wagenmakers, 2018) some propose the use of flexible *BF* thresholds (e.g., 'negotiable' thresholds), decided upon on a case-by-case basis (see Dienes, 2016; Dienes & Mclatchie, 2018 for similar discussions).

Further, it should be noted that, when designing the models for Bayesian studies, the expected effect size (reflected in the scale parameter *r* of the Cauchy prior distribution in the simulations conducted here), should concern the minimum effect size of interest (Dienes, 2019). For example, Brysbaert (2019) discusses that a common minimum effect size of interest in psychology is a standardized difference of $d = 0.4$. According to the results from

the simulations, researchers interested in this difference (d = 0.4) as the minimum effect size of interest, should expect to reach a threshold of $BF_{10} \geq 3$ if they use a sample size n = 30 (where n is the total sample size for within-subject designs, and group sample size for between-subject designs) with a probability of 25% if they assume that the variability of their data could lay anywhere between 0-4 standard deviations ($0 \leq sd \leq 4$; Table 1), a probability of 59% if they assume that the data will have low variability ($0 \leq sd \leq 1$; Table 3), 29% probability for data with expected moderate variability ($1 \leq sd \leq 2$; Table 5), and 19% probability for data with high variability ($2 \leq sd \leq 3$; Table 7). If a null effect is true and no differences exist in the comparisons of the two means, assuming a similar minimum effect size of interest d = 0.4, then a $BF_{10} \leq 1/3$ with sample size n = 30, is expected with 30% probability for data with unknown ($0 \leq sd \leq 4$; Table 2), low ($0 \leq sd \leq 1$; Table 4) moderate ($1 \leq sd \leq 2$; Table 6), and high ($2 \leq sd \leq 3$; Table 8) variability.

Also, it should be mentioned that the probabilities provided by the reference tables, do not correspond to the probability of successfully surpassing a *BF* threshold for a given experimental design. It is important to distinguish that the minimum effect size of interest discussed above, will in most cases differ to the observed difference in an experiment (for a similar argument see Dienes, 2019). Thus, the probabilities provided by the Tables should only be used as reference, and only when similar assumptions to the ones employed here can be justified (Lakens, 2022). To illustrate this, consider the study by Taikapi et al. (2022), who designed a within-subject experiment with sample size n = 15 to test the effects of a specific brain stimulation protocol on cortical excitability before and after stimulation. The authors were interested in testing for a minimum effect size of d = 0.7, and employed a Bayesian paired *t*-test, which reflected this prior assumption, for cortical excitability before brain stimulation and cortical excitability 45 minutes after brain stimulation. The *t*-test yielded a BF_{10} = 34.87. In general, brain stimulation studies are expected to result in data with high

variability (Phylactou et al., 2022). Therefore, assuming high variability, we turn to Table 7 for reference. According to Table 7 a minimum effect size of $d = 0.7$, a $BF_{10} \geq 3$ with such small sample size ($n < 20$) is less than 30% probable. However, this probability is only true if the observed effect size in the experiment is approximately $d \approx 0.7$. In such case, if the experiment is repeated 10000 times, and the true effect size is $d \approx 0.7$, then only 30% of the times a threshold of $BF_{10} \geq 3$ will be reached. However, in the experiment of Traikapi et al., (2022), the difference before and after stimulation were consistent and large ($d = 1.1$), which eventually resulted in a higher *BF*. This is also discussed in previous work, which illustrated that the resulting *BF* in a given experiment relies heavily on the observed data and that, usually, assumptions reflected in the prior distributions have little impact on the resulting *BF* (Schönbrodt et al., 2017; van Ravenzwaaij & Etz, 2021; Wagenmakers et al., 2010).

Overall, the reference tables are intended to provide various heuristics for the required sample size, when designing experiments interested in testing differences between two means. These heuristics are summarized in eight tables to cover various assumptions that are common in the social sciences. Specifically, the assumptions include testing for evidence under both the null and alternative hypotheses, a wide range of effect sizes, numerous sample sizes, different data variability expectations, and various *BF* thresholds. Even though the tables were generated to cover these common assumptions, social scientists should always carefully design their studies and justify their sample size according to prior assumptions that are most relevant to their aims, on a case-by-case basis (Dienes, 2016; Dienes & Mclatchie, 2018; Fu et al., 2021; Schönbrodt et al., 2017; Schönbrodt & Wagenmakers, 2018).

Concluding Remarks

Psychologists and social scientists cannot always rely on the updatable properties (Schönbrodt & Wagenmakers, 2018) of the *BF* when designing experiments. Often, they are required to decide upon and justify an anticipated sample size a-priori. Unfortunately, within

a Bayesian framework, such justification depends on data simulations (Fu et al., 2021; van Ravenzwaaij & Etz, 2021; see also Kruschke, 2013; Schönbrodt et al., 2017; Schönbrodt & Wagenmakers, 2018), which can be complicated, timely, and difficult to implement. Here, we computed such data simulations for various assumptions that are common in the social sciences, to help researchers and other stakeholders (e.g., reviewers, funders, etc.) when deciding on the expected sample size for a study planning to compare two means using the Bayesian *t*-test. We summarized these simulations in eight reference Tables, which can be used as heuristics, to estimate the probability of reaching a pre-determined *BF* threshold, given various experimental designs.

Code Availability

The python scripts used to generate the reference tables are available at the open science framework repository through https://osf.io/md6hr/?view_only=29c6ff272d3641e8b0bbf81a4a25dfc0.

Conflict of Interest

The authors confirm that they have no conflict of interest to declare.

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Figures

Figure 1: Visualization of the Monte Carlo simulations used to estimate the probability of reaching a Bayes Factor threshold.

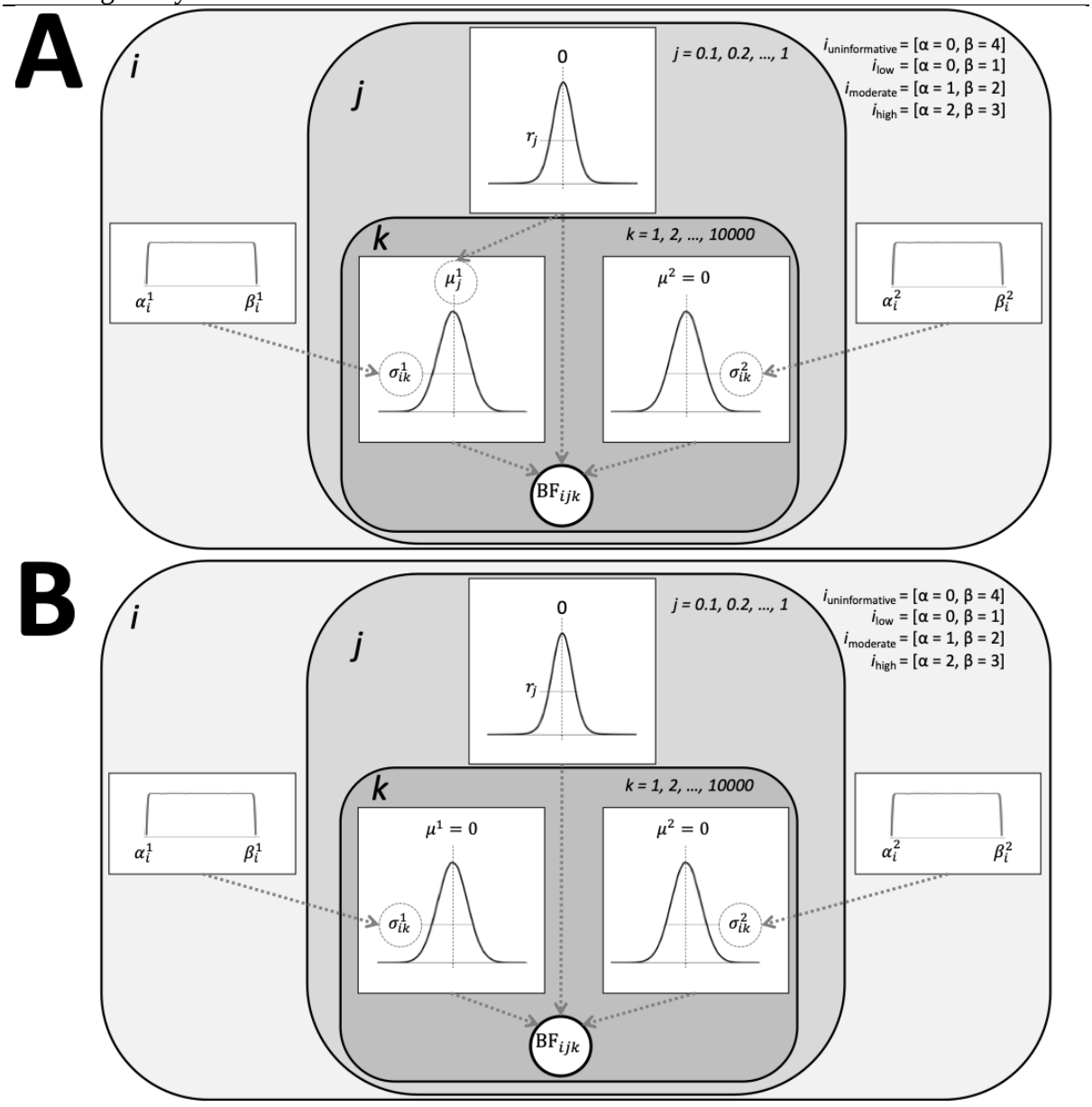


Fig1. The Monte Carlo simulations conducted to estimate the probability of surpassing a predefined Bayes Factor threshold. **(A)** The probability of surpassing the Bayes Factor threshold, assuming that a specific effect size is true, thus providing evidence in favor of the alternative hypothesis. The effect size expectations were reflected by the scale of a Cauchy distribution centered on 0 for *j* effect sizes (*j* = 0.1, 0.2, …, 1). Bayes Factors were calculated through a Bayesian paired *t*-test, using the same Cauchy distribution as the prior distribution of the *t*-test, which compared observations from two normal distributions. The first distribution had its mean drawn from the Cauchy distribution in each simulation ($\mu^1 \sim$ Cauchy(0, r_j)), while the second had a mean equal to 0 ($\mu^2 = 0$).. The standard deviation of each normal distribution was drawn by a separate uniform distribution. The parameters (α, β) of the uniform distributions were set according to *i* = 4 different assumptions, to reflect uninformative (α = 0, β = 4), low (α = 0, β = 1), moderate (α = 1, β = 2), and high (α = 2, β = 3) data variability. Each simulation was repeated *k* = 10000 times. **(B)** The probability of exceeding the Bayes Factor threshold, assuming that a null effect is true, therefore providing evidence in favor of the null hypothesis. The simulation procedure was the same as in (A) however, in (B) both normal distribution means were equal to 0 ($\mu^1 = \mu^2 = 0$).

Tables

Table 1. Estimated probability of finding a BF_{10} above predetermined thresholds (evidence for H_1), across different sample sizes and expected effect sizes, for data with unknown variability (between 0 and 4 standard deviations).

		Percentage (%) of BF_{10} above threshold for sample size (*n*):							
Cauchy prior scale (*r*)	**Threshold (BF_{10})**	**5**	**10**	**20**	**30**	**40**	**50**	**75**	**100**
0.1	2	3	5	8	10	12	13	14	16
	3	1	3	6	7	8	10	12	13
	5	1	2	5	6	7	7	10	12
	6	1	2	4	5	7	8	9	11
	10	0	2	4	5	6	7	8	9
0.2	2	8	11	15	18	20	22	24	27
	3	5	9	12	15	17	20	22	24
	5	3	6	10	13	15	16	19	22
	6	3	5	9	11	13	15	19	21
	10	2	5	8	10	12	13	17	18
0.3	2	12	16	21	25	27	28	32	34
	3	8	13	18	21	23	25	28	31
	5	5	9	15	17	21	23	26	29
	6	5	9	14	16	20	21	25	28
	10	3	8	12	16	17	19	23	25
0.4	2	15	20	25	29	32	33	37	40
	3	12	16	22	25	28	30	34	38
	5	7	13	19	22	24	27	31	34
	6	7	12	19	22	24	26	31	34
	10	5	10	16	19	22	25	29	32
0.5	2	18	24	29	32	36	38	42	44
	3	13	20	26	30	33	34	40	43
	5	10	16	22	27	29	33	36	39
	6	9	15	22	25	29	32	36	39
	10	6	13	20	23	27	29	33	37
0.6	2	21	26	33	37	39	41	46	48
	3	16	23	29	32	36	38	43	46
	5	11	18	26	30	33	36	41	43
	6	11	18	24	29	32	35	40	44
	10	9	16	22	27	29	33	38	41
0.7	2	23	28	34	39	42	44	49	53
	3	19	25	32	37	40	42	46	50
	5	14	21	27	33	37	39	44	47
	6	13	20	28	32	36	39	43	47
	10	10	18	26	30	34	36	41	46
0.8	2	25	31	38	42	45	47	52	55
	3	20	28	33	39	42	45	49	54
	5	16	24	32	36	38	42	47	51
	6	15	23	31	36	39	42	46	51
	10	12	20	28	32	37	40	45	48
0.9	2	27	33	39	44	47	50	54	58
	3	22	29	37	42	45	47	52	55
	5	18	26	34	39	42	45	50	54
	6	16	25	34	39	41	45	50	53
	10	14	21	31	36	39	43	48	52
1	2	28	34	42	45	50	52	58	60
	3	25	31	39	44	47	49	56	57
	5	20	28	36	41	45	48	53	56
	6	19	27	36	41	44	46	51	55
	10	15	24	33	38	42	45	51	55

Note: Sample size (*n*) refers to total *n* in repeated measure designs, but to group size *n* in independent measure designs (thus total sample size is *n * 2* for independent measures).

Table 2. Estimated probability of finding a BF_{10} below predetermined thresholds (evidence for H_0), across different sample sizes and expected effect sizes, for data with unknown variability (between 0 and 4 standard deviations).

		Percentage (%) of BF_{10} below threshold for sample size (*n*):							
Cauchy prior scale (*r*)	**Threshold (BF_{10})**	**5**	**10**	**20**	**30**	**40**	**50**	**75**	**100**
0.1	1/2	0	0	0	0	0	0	0	0
	1/3	0	0	0	0	0	0	0	0
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.2	1/2	0	0	0	10	40	51	62	68
	1/3	0	0	0	0	0	0	0	0
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.3	1/2	0	0	45	61	66	70	76	79
	1/3	0	0	0	0	0	24	50	60
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.4	1/2	0	37	64	71	75	79	82	85
	1/3	0	0	0	30	48	56	67	74
	1/5	0	0	0	0	0	0	0	27
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.5	1/2	16	54	71	77	80	83	86	88
	1/3	0	0	33	53	62	69	75	79
	1/5	0	0	0	0	0	0	40	54
	1/6	0	0	0	0	0	0	0	31
	1/10	0	0	0	0	0	0	0	0
0.6	1/2	40	63	76	80	84	86	88	90
	1/3	0	0	51	64	70	74	80	83
	1/5	0	0	0	0	12	37	58	65
	1/6	0	0	0	0	0	0	37	53
	1/10	0	0	0	0	0	0	0	0
0.7	1/2	50	69	80	83	86	87	90	92
	1/3	0	30	61	71	75	79	83	86
	1/5	0	0	0	16	42	53	66	72
	1/6	0	0	0	0	0	30	53	62
	1/10	0	0	0	0	0	0	0	0
0.8	1/2	56	73	81	86	88	89	92	92
	1/3	0	46	67	75	80	82	86	88
	1/5	0	0	0	42	55	63	71	77
	1/6	0	0	0	0	33	47	61	70
	1/10	0	0	0	0	0	0	0	17
0.9	1/2	62	76	84	87	89	91	92	93
	1/3	0	54	72	78	82	84	87	90
	1/5	0	0	28	52	61	68	76	80
	1/6	0	0	0	27	48	56	68	73
	1/10	0	0	0	0	0	0	0	40
1	1/2	66	78	85	88	90	91	93	94
	1/3	29	59	75	82	83	86	89	91
	1/5	0	0	42	60	66	72	79	82
	1/6	0	0	0	42	56	63	72	77
	1/10	0	0	0	0	0	0	34	51

Note: Sample size (*n*) refers to total *n* in repeated measure designs, but to group size *n* in independent measure designs (thus total sample size is *n * 2* for independent measures).

Table 3. Estimated probability of finding a BF_{10} above predetermined thresholds (evidence for H_1), across different sample sizes and expected effect sizes, for data with low variability (up to 1 standard deviation).

		Percentage (%) of BF_{10} above threshold for sample size (*n*):							
Cauchy prior scale (*r*)	**Threshold (BF_{10})**	**5**	**10**	**20**	**30**	**40**	**50**	**75**	**100**
0.1	2	8	15	23	27	31	34	38	42
	3	5	12	18	23	26	30	35	38
	5	4	9	15	20	22	26	31	34
	6	3	9	15	19	23	25	30	33
	10	2	8	13	17	20	22	28	30
0.2	2	20	30	39	45	49	52	56	59
	3	15	25	35	41	44	48	54	57
	5	11	21	31	37	41	43	49	53
	6	10	20	30	35	40	43	49	53
	10	7	17	27	32	37	39	45	50
0.3	2	28	40	50	55	59	61	66	69
	3	22	35	46	51	55	58	63	66
	5	18	30	41	47	52	56	60	64
	6	16	29	40	46	51	54	60	63
	10	13	25	37	44	47	52	58	61
0.4	2	36	47	56	61	65	68	72	75
	3	31	42	53	59	63	65	70	72
	5	23	37	49	55	59	62	68	71
	6	22	36	49	54	58	61	67	69
	10	18	32	45	51	56	59	64	68
0.5	2	42	53	62	67	70	72	76	78
	3	35	48	59	63	68	69	75	76
	5	29	45	55	61	65	67	73	75
	6	27	42	54	60	64	67	72	75
	10	22	38	52	57	62	66	70	74
0.6	2	47	58	66	71	74	76	79	81
	3	41	53	63	67	71	73	77	79
	5	34	49	59	66	68	71	77	78
	6	33	48	59	65	68	71	75	78
	10	27	45	56	63	65	70	74	76
0.7	2	51	61	68	74	76	77	81	84
	3	45	57	67	71	74	76	79	82
	5	39	54	63	68	73	75	79	81
	6	37	52	62	68	72	74	79	80
	10	32	48	60	67	69	72	76	79
0.8	2	54	64	72	76	78	80	83	85
	3	48	61	69	73	77	78	82	84
	5	43	57	67	72	75	77	81	82
	6	41	55	66	72	75	77	79	83
	10	36	53	63	69	73	76	80	81
0.9	2	57	67	74	78	80	82	84	85
	3	52	63	72	76	79	81	83	85
	5	46	59	70	74	77	79	82	84
	6	44	58	69	74	76	78	82	84
	10	39	55	66	72	75	78	81	83
1	2	59	69	76	79	82	83	86	87
	3	55	65	74	78	80	82	85	86
	5	49	62	71	76	79	81	84	86
	6	47	61	71	76	78	79	83	85
	10	42	59	70	74	77	79	82	85

Note: Sample size (*n*) refers to total *n* in repeated measure designs, but to group size *n* in independent measure designs (thus total sample size is *n * 2* for independent measures).

Table 4. Estimated probability of finding a BF_{10} below predetermined thresholds (evidence for H_0), across different sample sizes and expected effect sizes, for data with low variability (up to 1 standard deviation).

		Percentage (%) of BF_{10} below threshold for sample size (*n*):							
Cauchy prior scale (*r*)	**Threshold (BF_{10})**	**5**	**10**	**20**	**30**	**40**	**50**	**75**	**100**
0.1	1/2	0	0	0	0	0	0	0	0
	1/3	0	0	0	0	0	0	0	0
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.2	1/2	0	0	0	10	40	51	62	68
	1/3	0	0	0	0	0	0	0	0
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.3	1/2	0	0	45	61	66	70	76	79
	1/3	0	0	0	0	0	24	50	60
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.4	1/2	0	37	64	71	75	79	82	85
	1/3	0	0	0	30	48	56	67	74
	1/5	0	0	0	0	0	0	0	27
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.5	1/2	16	54	71	77	80	83	86	88
	1/3	0	0	33	53	62	69	75	79
	1/5	0	0	0	0	0	0	40	54
	1/6	0	0	0	0	0	0	0	31
	1/10	0	0	0	0	0	0	0	0
0.6	1/2	40	63	76	80	84	86	88	90
	1/3	0	0	51	64	70	74	80	83
	1/5	0	0	0	0	12	37	58	65
	1/6	0	0	0	0	0	0	37	53
	1/10	0	0	0	0	0	0	0	0
0.7	1/2	50	69	80	83	86	87	90	92
	1/3	0	30	61	71	75	79	83	86
	1/5	0	0	0	16	42	53	66	72
	1/6	0	0	0	0	0	30	53	62
	1/10	0	0	0	0	0	0	0	0
0.8	1/2	56	73	81	86	88	89	92	92
	1/3	0	46	67	75	80	82	86	88
	1/5	0	0	0	42	55	63	71	77
	1/6	0	0	0	0	33	47	61	70
	1/10	0	0	0	0	0	0	0	17
0.9	1/2	62	76	84	87	89	91	92	93
	1/3	0	54	72	78	82	84	87	90
	1/5	0	0	28	52	61	68	76	80
	1/6	0	0	0	27	48	56	68	73
	1/10	0	0	0	0	0	0	0	40
1	1/2	66	78	85	88	90	91	93	94
	1/3	29	59	75	82	83	86	89	91
	1/5	0	0	42	60	66	72	79	82
	1/6	0	0	0	42	56	63	72	77
	1/10	0	0	0	0	0	0	34	51

Note: Sample size (*n*) refers to total *n* in repeated measure designs, but to group size *n* in independent measure designs (thus total sample size is *n * 2* for independent measures).

Table 5. Estimated probability of finding a BF_{10} above predetermined thresholds (evidence for H_1), across different sample sizes and expected effect sizes, for data with moderate variability (between 1 and 2 standard deviations).

		Percentage (%) of BF_{10} above threshold for sample size (*n*):							
Cauchy prior scale (*r*)	**Threshold (BF_{10})**	**5**	**10**	**20**	**30**	**40**	**50**	**75**	**100**
0.1	2	3	6	9	11	13	14	16	18
	3	1	4	6	8	10	11	14	15
	5	1	3	5	6	7	9	11	13
	6	0	2	4	6	8	9	10	12
	10	0	2	4	5	6	7	9	11
0.2	2	8	13	18	21	24	25	29	31
	3	5	9	14	18	19	22	25	28
	5	3	7	11	15	17	18	21	25
	6	3	6	11	13	15	18	21	25
	10	2	5	9	11	14	15	19	21
0.3	2	13	18	24	28	31	32	37	41
	3	8	15	20	24	27	30	34	37
	5	6	11	17	20	24	27	31	34
	6	6	10	16	19	23	25	30	33
	10	4	9	15	18	19	23	28	31
0.4	2	17	23	29	33	37	39	44	47
	3	13	19	26	29	33	35	41	45
	5	8	15	22	26	29	32	38	41
	6	7	14	22	26	29	31	37	40
	10	6	12	18	22	27	30	34	38
0.5	2	20	27	34	38	42	44	49	52
	3	15	23	30	35	39	40	47	50
	5	11	19	27	32	35	38	43	46
	6	10	18	26	30	34	38	43	46
	10	7	15	23	27	32	35	40	44
0.6	2	23	30	38	43	46	49	54	56
	3	18	26	33	39	42	44	50	54
	5	13	22	30	36	39	42	49	51
	6	13	21	29	35	39	41	47	52
	10	10	19	27	32	35	39	44	48
0.7	2	26	33	40	46	49	52	57	61
	3	21	29	38	43	47	50	54	58
	5	16	24	33	39	44	46	52	56
	6	16	24	33	38	43	46	52	55
	10	12	21	31	37	40	43	49	53
0.8	2	28	36	44	49	52	55	60	63
	3	22	32	40	45	49	53	57	62
	5	19	28	38	42	46	51	55	60
	6	17	27	36	43	46	50	54	58
	10	14	24	33	39	43	47	54	56
0.9	2	31	38	46	52	55	58	63	66
	3	26	34	43	49	52	55	60	64
	5	21	30	41	46	50	53	59	62
	6	19	29	40	46	49	52	58	62
	10	16	26	36	43	47	51	56	60
1	2	32	40	49	53	58	60	66	68
	3	29	37	46	51	55	58	64	66
	5	23	33	43	49	53	56	62	65
	6	22	32	42	48	52	54	60	63
	10	18	29	39	45	49	53	59	63

Note: Sample size (*n*) refers to total *n* in repeated measure designs, but to group size *n* in independent measure designs (thus total sample size is *n * 2* for independent measures).

Table 6. Estimated probability of finding a BF_{10} below predetermined thresholds (evidence for H_0), across different sample sizes and expected effect sizes, for data with moderate variability (between 1 and 2 standard deviations).

		Percentage (%) of BF_{10} below threshold for sample size (n):							
Cauchy prior scale (r)	**Threshold (BF_{10})**	**5**	**10**	**20**	**30**	**40**	**50**	**75**	**100**
0.1	1/2	0	0	0	0	0	0	0	0
	1/3	0	0	0	0	0	0	0	0
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.2	1/2	0	0	0	9	40	51	63	69
	1/3	0	0	0	0	0	0	0	0
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.3	1/2	0	0	45	61	66	70	76	79
	1/3	0	0	0	0	0	24	50	59
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.4	1/2	0	38	63	71	76	79	82	86
	1/3	0	0	0	30	48	56	67	73
	1/5	0	0	0	0	0	0	0	28
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.5	1/2	16	55	71	77	80	83	86	88
	1/3	0	0	33	53	62	69	75	79
	1/5	0	0	0	0	0	0	40	54
	1/6	0	0	0	0	0	0	0	31
	1/10	0	0	0	0	0	0	0	0
0.6	1/2	39	63	76	81	84	86	89	90
	1/3	0	0	51	64	70	74	80	83
	1/5	0	0	0	0	12	37	58	66
	1/6	0	0	0	0	0	0	37	52
	1/10	0	0	0	0	0	0	0	0
0.7	1/2	50	69	80	83	86	87	90	91
	1/3	0	30	61	71	75	79	83	86
	1/5	0	0	0	16	42	53	66	72
	1/6	0	0	0	0	0	30	53	62
	1/10	0	0	0	0	0	0	0	0
0.8	1/2	57	73	81	86	88	89	91	93
	1/3	0	46	67	75	79	82	86	88
	1/5	0	0	0	42	55	63	71	77
	1/6	0	0	0	0	32	47	61	70
	1/10	0	0	0	0	0	0	0	17
0.9	1/2	62	77	84	87	89	91	92	93
	1/3	0	54	71	78	82	84	87	90
	1/5	0	0	29	52	61	68	76	80
	1/6	0	0	0	28	48	56	68	73
	1/10	0	0	0	0	0	0	0	40
1	1/2	65	78	86	88	90	91	93	94
	1/3	29	60	75	81	84	86	89	91
	1/5	0	0	42	60	66	72	79	83
	1/6	0	0	0	42	56	63	71	78
	1/10	0	0	0	0	0	0	34	51

Note: Sample size (n) refers to total n in repeated measure designs, but to group size n in independent measure designs (thus total sample size is $n * 2$ for independent measures).

Table 7. Estimated probability of finding a BF_{10} above predetermined thresholds (evidence for H_1), across different sample sizes and expected effect sizes, for data with high variability (between 2 and 3 standard deviations).

		Percentage (%) of BF_{10} above threshold for sample size (*n*):							
Cauchy prior scale (*r*)	**Threshold (BF_{10})**	**5**	**10**	**20**	**30**	**40**	**50**	**75**	**100**
0.1	2	2	4	6	7	9	10	11	13
	3	1	2	4	5	6	7	9	10
	5	0	2	3	4	5	5	7	8
	6	0	1	2	4	5	5	6	7
	10	0	1	2	3	4	4	6	6
0.2	2	6	9	12	14	16	17	20	21
	3	3	6	9	11	13	15	16	19
	5	2	4	7	9	11	12	14	17
	6	2	3	7	8	10	11	14	16
	10	1	3	5	7	9	9	12	13
0.3	2	9	13	17	20	22	22	25	28
	3	6	10	13	16	18	20	22	25
	5	3	7	11	13	16	18	20	23
	6	3	7	10	12	15	16	19	22
	10	2	5	9	11	12	15	18	20
0.4	2	12	15	20	23	26	27	31	34
	3	9	12	17	19	22	24	28	32
	5	5	10	14	17	19	21	26	28
	6	5	9	14	17	19	20	25	27
	10	3	7	12	14	17	19	23	25
0.5	2	15	19	23	26	30	31	36	38
	3	10	15	20	24	27	28	33	36
	5	7	12	17	21	23	26	29	32
	6	7	11	16	20	23	25	29	32
	10	4	10	15	17	21	23	27	31
0.6	2	17	21	26	31	33	35	39	42
	3	12	17	23	26	29	31	37	39
	5	9	14	20	24	26	29	34	36
	6	8	14	19	23	26	28	33	37
	10	6	12	17	21	23	26	31	34
0.7	2	19	23	29	32	35	38	43	47
	3	15	20	26	30	33	36	40	44
	5	11	16	21	26	31	32	37	41
	6	10	16	22	25	29	32	37	40
	10	7	13	20	24	28	29	34	39
0.8	2	21	25	31	35	38	41	46	49
	3	16	22	27	33	35	38	43	47
	5	12	19	26	29	32	36	40	45
	6	11	17	24	29	32	35	39	44
	10	8	15	22	26	29	33	38	41
0.9	2	22	26	33	38	41	44	48	52
	3	17	23	30	35	38	40	46	49
	5	14	20	28	32	36	39	44	47
	6	12	19	27	32	35	38	43	46
	10	10	16	25	29	32	36	41	45
1	2	23	29	35	39	44	46	51	54
	3	20	25	33	37	41	43	50	52
	5	16	22	29	34	39	41	47	50
	6	14	21	29	34	37	39	45	49
	10	11	19	26	31	35	38	44	48

Note: Sample size (*n*) refers to total *n* in repeated measure designs, but to group size *n* in independent measure designs (thus total sample size is *n * 2* for independent measures).

Table 8. Estimated probability of finding a BF_{10} below predetermined thresholds (evidence for H_0), across different sample sizes and expected effect sizes, for data with high variability (between 2 and 3 standard deviations).

		Percentage (%) of BF_{10} below threshold for sample size (*n*):							
Cauchy prior scale (*r*)	**Threshold (BF_{10})**	**5**	**10**	**20**	**30**	**40**	**50**	**75**	**100**
0.1	1/2	0	0	0	0	0	0	0	0
	1/3	0	0	0	0	0	0	0	0
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.2	1/2	0	0	0	10	40	51	63	69
	1/3	0	0	0	0	0	0	0	0
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.3	1/2	0	0	45	60	66	70	76	79
	1/3	0	0	0	0	0	23	50	59
	1/5	0	0	0	0	0	0	0	0
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.4	1/2	0	38	63	71	76	79	83	85
	1/3	0	0	0	30	48	56	67	73
	1/5	0	0	0	0	0	0	0	28
	1/6	0	0	0	0	0	0	0	0
	1/10	0	0	0	0	0	0	0	0
0.5	1/2	16	55	71	77	80	83	86	88
	1/3	0	0	33	53	62	69	75	79
	1/5	0	0	0	0	0	0	40	54
	1/6	0	0	0	0	0	0	0	32
	1/10	0	0	0	0	0	0	0	0
0.6	1/2	39	63	76	81	84	86	89	90
	1/3	0	0	51	64	70	74	80	83
	1/5	0	0	0	0	12	37	58	66
	1/6	0	0	0	0	0	0	37	52
	1/10	0	0	0	0	0	0	0	0
0.7	1/2	50	69	80	83	86	87	90	91
	1/3	0	30	61	70	75	79	83	86
	1/5	0	0	0	17	42	53	66	72
	1/6	0	0	0	0	0	30	53	62
	1/10	0	0	0	0	0	0	0	0
0.8	1/2	57	73	81	86	88	89	91	92
	1/3	0	45	67	75	80	82	86	88
	1/5	0	0	0	41	55	62	71	77
	1/6	0	0	0	0	32	47	61	70
	1/10	0	0	0	0	0	0	0	17
0.9	1/2	62	76	84	87	89	90	92	93
	1/3	0	54	71	78	82	84	87	90
	1/5	0	0	29	52	61	68	76	80
	1/6	0	0	0	28	48	56	68	73
	1/10	0	0	0	0	0	0	0	40
1	1/2	65	77	85	88	90	91	93	94
	1/3	29	60	75	81	84	85	89	91
	1/5	0	0	43	60	67	72	79	83
	1/6	0	0	0	42	56	63	71	78
	1/10	0	0	0	0	0	0	33	51

Note: Sample size (*n*) refers to total *n* in repeated measure designs, but to group size *n* in independent measure designs (thus total sample size is *n * 2* for independent measures).